

SPATIAL REGULARIZATION IN FEDERATED LEARNING: PREDICTING TEMPERATURE VIA NETWORK CONSENSUS

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ABSTRACT

In real-world Federated Learning (FL) applications, individual clients—such as weather stations—often possess limited, noisy, or geographically sparse data, making them susceptible to overfitting. This project tackles the challenge of data scarcity by connecting Finnish Meteorological Institute (FMI) weather stations into a spatial graph. By leveraging the Graph Total Variation Minimization (GTVMIn) framework, we impose a structural prior on the learning process: neighboring stations should learn structurally similar local models. We formulate the weather prediction task as a distributed optimization problem, solved via a block-wise minimization method. Finally, we establish an experimental setup to evaluate the proposed FL framework against local-only and global shared model baselines.

Index Terms— Federated learning, networks, variations, weather prediction, machine learning

1. INTRODUCTION

In real-world Federated Learning (FL) [2] applications, individual clients—in this case, weather stations—often possess limited, noisy, or geographically sparse data. When a single weather station attempts to train a machine learning model using only its own small dataset, the model is highly susceptible to overfitting and typically generalizes poorly to unseen data.

This project tackles the challenge of data scarcity by connecting weather stations into a spatial graph. By leveraging the Graph Total Variation Minimization (GTVMIn) framework [1], we impose a structural prior on the learning process: neighboring stations (nodes connected by edges) should learn local models that are structurally similar. This injects the physical domain knowledge that weather patterns are spatially continuous, allowing individual stations to collaborate without centralizing their raw data.

The remainder of this report expands strictly on this premise. Section 2 formalizes the network setup and local modeling task. Section 3 derives the GTVMIn methodology and local update operators. Section 4 details the planned numerical evaluation, and Section 5 discusses expected conclusions and limitations.

2. PROBLEM FORMULATION

This section formally maps the physical weather prediction problem into the Federated Learning framework, defining the data structures, network configuration, and the learning objective for individual stations. The network is modeled over an empirical graph $\mathcal{G} = (\mathcal{V}, \mathcal{E}, A)$.

Data and Task: The predictive task is estimating Air Temperature based on local meteorological factors. The dataset spans January 1, 2024 to February 1, 2024. Features ($\mathbf{x}$) are derived from available meteorological readings including Cloud amount, Dew-point temperature, Gust speed, Horizontal visibility, Precipitation amount, Precipitation intensity, Present weather (auto), Pressure (msl), Relative humidity, Snow depth, Wind direction, and Wind speed. The data structure also records station name, FMI ID, spatial coordinates, timestamps, and measurement units. The label (y) is Air temperature. Each station i has a local dataset of m_i readings: $\mathcal{D}^{(i)} = \{(\mathbf{x}_1^{(i)}, y_1^{(i)}), \dots, (\mathbf{x}_{m_i}^{(i)}, y_{m_i}^{(i)})\}$.

Network Topology and Models: Each node $i \in \mathcal{V}$ represents an FMI weather station. Edges $\mathcal{E}$ are defined by the Delaunay Triangulation of the stations' geographic coordinates (longitude, latitude). The Delaunay Triangulation creates a non-overlapping mesh of triangles connecting the points such that no point lies inside the circumcircle of any triangle. This mathematical property is needed because it naturally connects adjacent neighbor stations without relying on arbitrary, hard-coded distance thresholds. Edge Weights $A_{i,j}$ are determined by the Haversine distance. For an edge between station i and j , $A_{i,j} = \max(0, 60 - \text{dist}(i, j))$. This ensures that only local climatic neighborhoods influence each other, degrading linearly to 0 at 60km.

Hypothesis Class and Loss: We use Linear Regression. For each station i , the model is parameterized by a weight vector $\mathbf{w}^{(i)}$, with predictions $\hat{y} = (\mathbf{w}^{(i)})^T \mathbf{x}$. The local loss $L_i(\mathbf{w}^{(i)})$ is the Mean Squared Error (MSE) computed on dataset $\mathcal{D}^{(i)}$.

3. METHODOLOGY

We formulate the FL application as a GTVMin optimization problem:

$$\min_{\mathbf{w}^{(1)}, \dots, \mathbf{w}^{(n)}} \sum_{i=1}^n L_i(\mathbf{w}^{(i)}) + \alpha \sum_{\{i,j\} \in \mathcal{E}} A_{i,j} \|\mathbf{w}^{(i)} - \mathbf{w}^{(j)}\|_2^2 \quad (1)$$

The first term ensures fidelity to local data, while the second term uses the ℓ_2 -norm squared as the variation measure $d(\mathbf{w}^{(i)}, \mathbf{w}^{(j)})$ to enforce consensus among neighboring models. Parameter α controls this trade-off.

We solve this via block-wise optimization. From the perspective of node i , holding neighbors fixed, let $d_i = \sum_{j \in \mathcal{N}^{(i)}} A_{i,j}$, and $\bar{\mathbf{w}}^{(i)} = \frac{1}{d_i} \sum_{j \in \mathcal{N}^{(i)}} A_{i,j} \mathbf{w}^{(j)}$. At iteration $t + 1$, node i updates its model by evaluating the block operator $\mathcal{F}^{(i)}$:

$$\mathcal{F}^{(i)} = \arg \min_{\mathbf{w}} \left(L_i(\mathbf{w}) + \alpha \cdot d_i \|\mathbf{w} - \bar{\mathbf{w}}^{(i,t)}\|_2^2 \right) \quad (2)$$

By taking the derivative with respect to $\mathbf{w}$ and setting it to zero, we arrive at the fixed-point equation:

$$\mathbf{w}^{(i,t+1)} = (X_i^T X_i + \alpha \cdot d_i I)^{-1} \left(X_i^T y_i + \alpha \cdot d_i \bar{\mathbf{w}}^{(i,t)} \right) \quad (3)$$

This mimics Ridge Regression centered around $\bar{\mathbf{w}}^{(i,t)}$. We realize this **explicitly** using a data augmentation trick with the `scikit-learn` [3] class `LinearRegression`. We augment local training features X_i with $\sqrt{\alpha d_i} I$, and labels y_i with $\sqrt{\alpha d_i} \bar{\mathbf{w}}^{(i,t)}$. We then call `.fit()` on the augmented datasets.

4. NUMERICAL EXPERIMENTS

This section will document the quantitative evaluation of the proposed framework. The time-series data will be chronologically divided into Train (60%), Validation (20%), and Test (20%) sets to intuitively respect temporal causality.

We evaluate and compare the FL GTVMin method against two structurally distinct baseline systems:

1. *Local-only Training* ($\alpha = 0$): Independent local models with zero edge weighting/communication.
2. *Global Shared Model* ($\alpha \rightarrow \infty$): A single model trained on universally pooled data. This model uses an inherently distinct graph construction (fully connected with infinite weights) and relies on its own separate L_2 (Ridge) structural modeling.

Hyperparameter tuning for the optimal α will be performed over a log-scale grid computing the MSE on the validation set.

Reproducibility & LLM Reimplementation: The finalized algorithms, equations, graph configurations, and explicit

chronological data selection (Jan 1, 2024 to Feb 1, 2024; 13 selected FMI measurement variables) detailed in earlier sections provide a fully-transparent blueprint enabling automated reimplementation using LLMs such as Claude Code.

5. CONCLUSION

Upon finalizing the numerical experiments, this section will interpret the trade-offs captured by the validation and test results. We expect the GTVMin model to outperform local models by mitigating data scarcity, and beat global models by preserving spatial microclimates. Potential limitations, such as assuming stationary station relations or strictly linear model capacities, will also be discussed to identify avenues for future improvement.

6. REFERENCES

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